

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT2 — DATA ANALYSIS PROJECT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Analysis (BL4)	Assessment:	Assessment Tool 2 – Data Analysis Project
Weightage:	40% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Analyze real datasets using ECDFs, skewed-distribution techniques, Q–Q plots, multi-distribution comparisons, and proportion visualizations to identify patterns and justify conclusions.(BL4)</i>		
Student Name:	Nivya Sri T	Reg. No.:	192424131
Section / Batch:	2024	Date:	13-08-2026

Code of Conduct

I Nivya Sri T (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nivya Sri T

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond construction: analyze the given data, identify the relevant pattern, and justify your conclusion.
- Show all supporting calculations (e.g., IQR, percentiles, angle calculations) where relevant.
- Where a question asks you to compare or recommend, state your reasoning explicitly — a correct chart choice without justification will not receive full marks.

Annexure A – DATA ANALYSIS PROJECT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Data Analysis Project

CO3 · BT4: Analyze ·

This rubric is used to assess student responses to the CO3-AT2 Data Analysis Project, in which students analyze a given dataset using ECDF, skewed-distribution, Q–Q plot, multi-distribution, or proportion-visualization techniques and justify their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant technique (ECDF, skewed-distribution/log transform, Q–Q plot, multi-distribution comparison, or proportion visualization) and correctly identifies which pattern applies to the given data.	Good understanding of the relevant technique, with only minor gaps in identifying the correct pattern.	Basic understanding, but with some misconceptions about which technique or pattern applies to the data.	Poor understanding of core concepts; misidentifies the pattern or technique required by the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, going beyond description to explain why the pattern occurs or what it implies.	Provides clear analysis with adequate reasoning tied to the data, for most parts.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	20%	Effectively applies analytical techniques to compare, justify, or recommend a conclusion that is well-supported by the data and any calculations shown.	Applies analytical techniques correctly, with only minor errors in reasoning or calculation.	Limited application; conclusions are only partially supported by the data or calculations.	Fails to apply appropriate analytical techniques; conclusions are unsupported or incorrect.
Organization	15%	The response is clearly structured, addressing every sub-part (a–b) in	The response is organized and addresses each sub-part, with	Basic structure; sub-parts are addressed but the response lacks	Poorly organized; the response is incoherent, and sub-parts are left

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		a logical sequence with coherent overall flow.	only minor issues in flow or clarity.	clarity or a logical sequence.	unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — pattern identification, calculations, and conclusions — are completely accurate and well-suited to the data.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Problem 3 — ECDF

a) ECDF + Median + 90th Percentile

Median

Middle value of 15 sorted scores $\rightarrow$ 8th value = 65

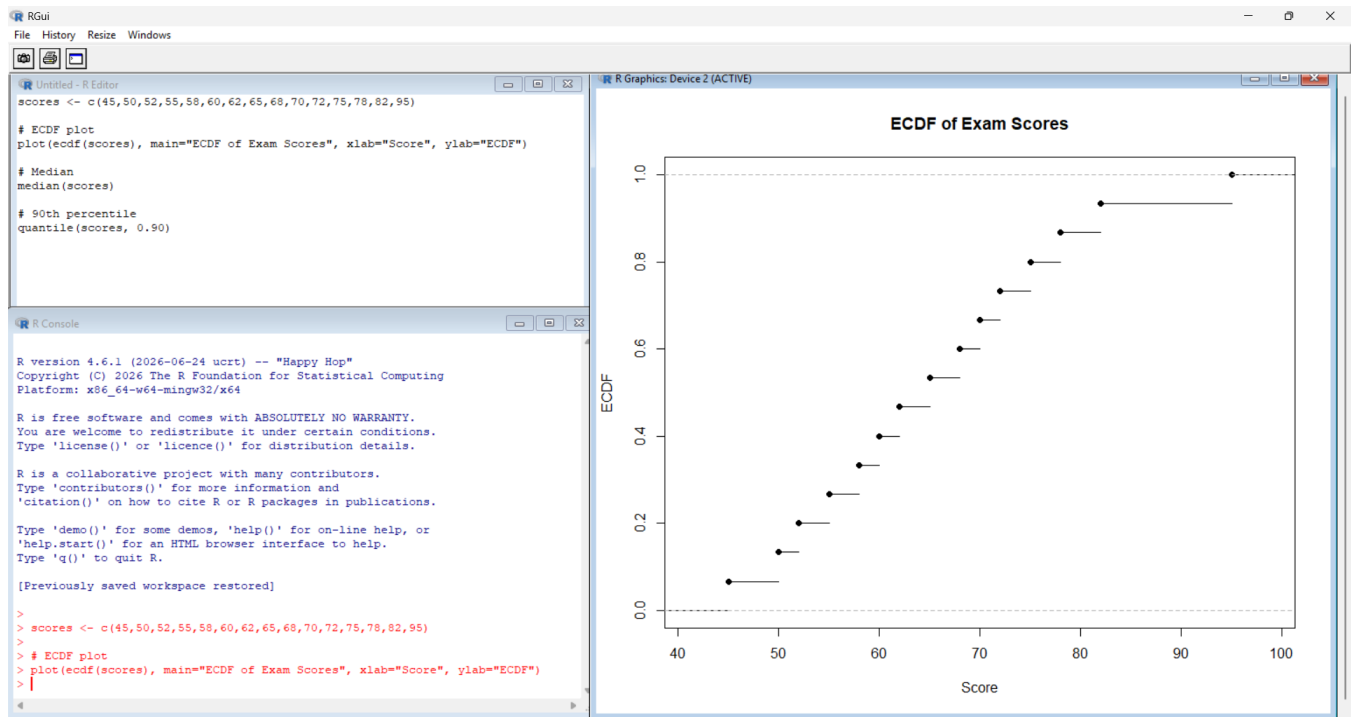
90th Percentile

90% of 15 = 13.5 $\rightarrow$ between 13th & 14th values $\rightarrow$ 13th = 78, 14th = 82 $\rightarrow \approx 82$

b) Outlier Analysis (Score = 95)

The ECDF curve will rise smoothly until the last jump. The final jump (from 82 $\rightarrow$ 95) is much larger than previous jumps, indicating:

- A long right tail
- 95 is far from the rest
- Appears as an outlier on ECDF because the curve flattens before a sudden jump



2. Problem 7 — Skewed Distributions & Log Transform

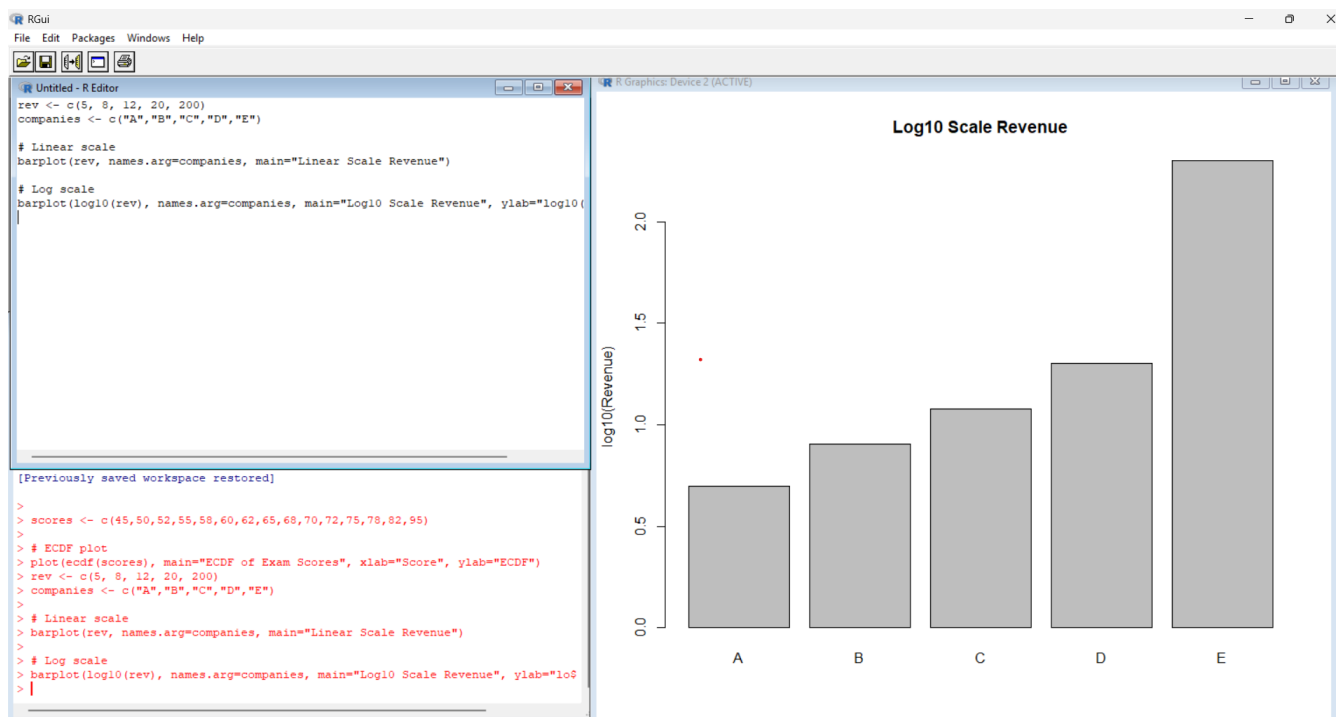
a) Why linear-scale bar chart is misleading?

- The value 200 is 10 $\times$ larger than the next highest (20).
- On a linear bar chart, the bar for 200 will dominate the plot.
- Smaller companies (5, 8, 12, 20) will appear almost flat.
- This hides meaningful differences among smaller values.

b) Better Axis Scale: Log Scale

Using log10 scale:

- Compresses large values
- Expands small values
- Shows proportional differences clearly
- Reveals pattern: revenues follow a heavy right-skew / power-law-like distribution



3. Problem 11 — Q-Q Plot

a) Interpretation

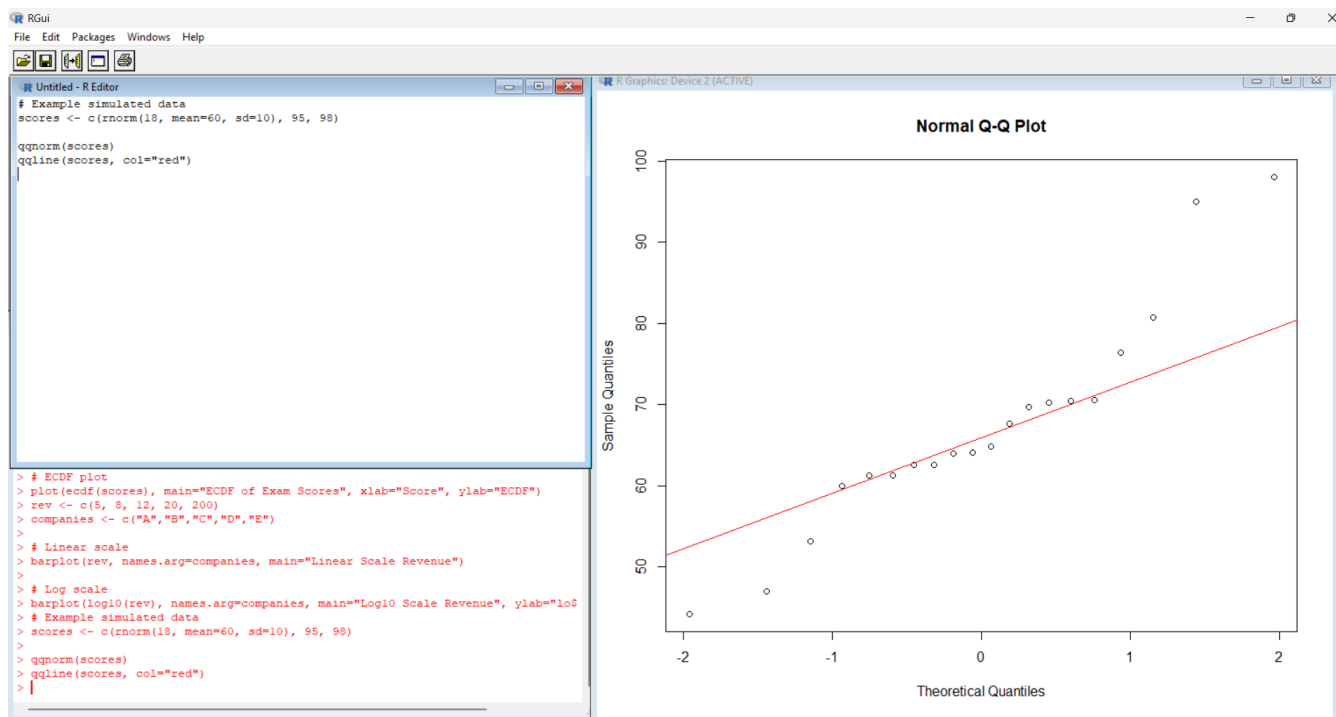
This pattern indicates:

- Right tail heavier than Normal
- High-end scores are larger than expected under normality
- Students at the top performed exceptionally well

b) Real-world Explanation

Possible reasons:

- Bonus marks
- Extra-credit questions
- Ceiling effect (students reaching near-max scores)
- Top performers pulling the tail upward



4. Problem 20 — Many Distributions at Once

a) Violin/Bean Plot Interpretation

Group 1 (Caffeine)

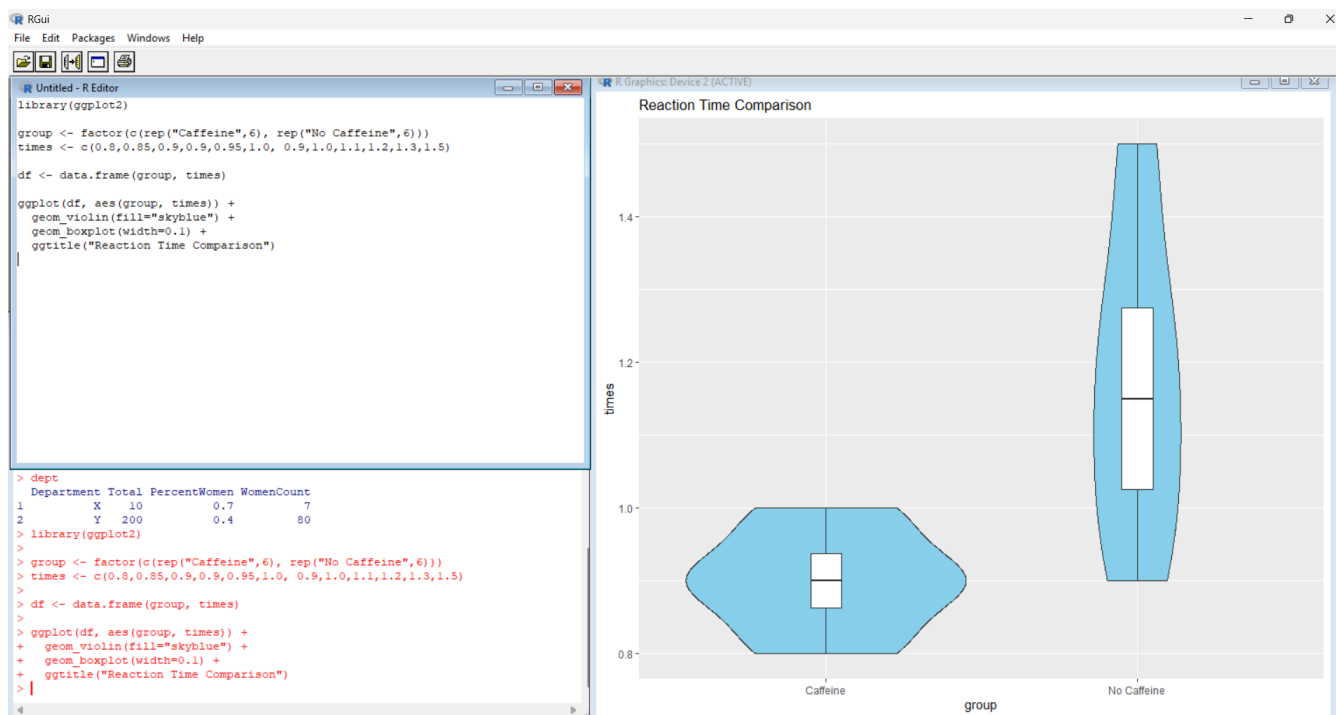
- Tight cluster around 0.85–1.0
- Narrow spread → consistent reaction times

Group 2 (No Caffeine)

- Wider spread (0.9–1.5)
- More variability → less consistent reaction times

b) Conclusion About Caffeine

- Caffeine reduces variability in reaction time
- Group 1 is more stable and consistent
- Even if averages are similar, spread shows caffeine improves reliability.



5. Problem 24 — Proportions & Misleading Visualizations

a) Why percentage-only reporting is misleading?

Percentages hide actual counts.

Example:

- X may have 70% women but only 10 employees → 7 women
- Y may have 40% women but 200 employees → 80 women

Thus, Y actually has more women, even though its percentage is lower.

Missing critical information: ✓ Total number of employees in each department.

b) Hypothetical Example

Department	Total Employees	% Women	Women Count
X	10	70%	7
Y	200	40%	80

→ Y employs more women (80) than X (7).

